

Cyclic Codes

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Description of Cyclic Codes

- Cyclic codes were invented by Prange in 1957.
- If the components of an n -tuple $\mathbf{v} = (v_0, v_1, \dots, v_{n-1})$ are cyclically shifted i places to the right, the resultant n -tuple would be

$$\mathbf{v}^{(i)} = (v_{n-i}, v_{n-i+1}, \dots, v_{n-1}, v_0, v_1, \dots, v_{n-i-1}).$$

- Cyclically shifting $\mathbf{v}$ i places to the right is equivalent to cyclically shifting $\mathbf{v}$ $n - i$ places to the left.
- An (n, k) linear code $\mathbf{C}$ is called a *cyclic code* if every cyclic shift of a codeword in $\mathbf{C}$ is also a codeword in $\mathbf{C}$.
- Code polynomial $\mathbf{v}(x)$ of the codeword $\mathbf{v}$ is defined as

$$\mathbf{v}(x) = v_0 + v_1x + \dots + v_{n-1}x^{n-1}.$$

- $\mathbf{v}^{(i)}(x) = x^i \mathbf{v}(x) \bmod x^n + 1.$

Proof: Multiplying $\mathbf{v}(x)$ by x^i , we obtain

$$x^i \mathbf{v}(x) = v_0 x^i + v_1 x^{i+1} + \cdots + v_{n-i-1} x^{n-1} + \cdots + v_{n-1} x^{n+i-1}.$$

Then we manipulate the equation into the following form:

$$\begin{aligned} x^i \mathbf{v}(x) &= v_{n-i} + v_{n-i+1} x + \cdots + v_{n-1} x^{i-1} + v_0 x^i + \cdots \\ &\quad + v_{n-i-1} x^{n-1} + v_{n-i}(x^n + 1) + v_{n-i+1} x(x^n + 1) \\ &\quad + \cdots + v_{n-1} x^{i-1}(x^n + 1) \\ &= \mathbf{q}(x)(x^n + 1) + \mathbf{v}^{(i)}(x), \end{aligned}$$

where $\mathbf{q}(x) = v_{n-i} + v_{n-i+1} x + \cdots + v_{n-1} x^{i-1}$.

- The nonzero code polynomial of minimum degree in a cyclic code $\mathbf{C}$ is unique.
- Let $\mathbf{g}(x) = g_0 + g_1 x + \cdots + g_{r-1} x^{r-1} + x^r$ be the nonzero code polynomial of minimum degree in an (n, k) cyclic code $\mathbf{C}$. Then the constant term g_0 must be equal to 1.

Proof: Suppose that $g_0 = 0$. Then

$$\begin{aligned}\mathbf{g}(x) &= g_1x + g_2x^2 + \cdots + g_{r-1}x^{r-1} + x^r \\ &= x(g_1 + g_2x + \cdots + g_{r-1}x^{r-2} + x^{r-1}).\end{aligned}$$

If we shift $\mathbf{g}(x)$ cyclically $n - 1$ places to the right (or one place to the left), we obtain a nonzero code polynomial,

$g_1 + g_2x + \cdots + g_{r-1}x^{r-2} + x^{r-1}$, which has a degree less than r . Contradiction.

A (7, 4) Cyclic Code Generated by $\mathbf{g}(x) = 1 + x + x^3$

Messages	codewords	Code Polynomials
(0 0 0 0)	0 0 0 0 0 0 0	$0 = 0 \cdot \mathbf{g}(x)$
(1 0 0 0)	1 1 0 1 0 0 0	$1 + x + x^3 = 1 \cdot \mathbf{g}(x)$
(0 1 0 0)	0 1 1 0 1 0 0	$x + x^2 + x^4 = x \cdot \mathbf{g}(x)$
(1 1 0 0)	1 0 1 1 1 0 0	$1 + x^2 + x^3 + x^4 = (1 + x) \cdot \mathbf{g}(x)$
(0 0 1 0)	0 0 1 1 0 1 0	$x^2 + x^3 + x^5 = x^2 \cdot \mathbf{g}(x)$
(1 0 1 0)	1 1 1 0 0 1 0	$1 + x + x^2 + x^5 = (1 + x^2) \cdot \mathbf{g}(x)$
(0 1 1 0)	0 1 0 1 1 1 0	$x + x^3 + x^4 + x^5 = (x + x^2) \cdot \mathbf{g}(x)$
(1 1 1 0)	1 0 0 0 1 1 0	$1 + x^4 + x^5 = (1 + x + x^2) \cdot \mathbf{g}(x)$
(0 0 0 1)	0 0 0 1 1 0 1	$x^3 + x^4 + x^6 = x^3 \cdot \mathbf{g}(x)$
(1 0 0 1)	1 1 0 0 1 0 1	$1 + x + x^4 + x^6 = (1 + x^3) \cdot \mathbf{g}(x)$
(0 1 0 1)	0 1 1 1 0 0 1	$x + x^2 + x^3 + x^6 = (x + x^3) \cdot \mathbf{g}(x)$
(1 1 0 1)	1 0 1 0 0 0 1	$1 + x^2 + x^6 = (1 + x + x^3) \cdot \mathbf{g}(x)$
(0 0 1 1)	0 0 1 0 1 1 1	$x^2 + x^4 + x^5 + x^6 = (x^2 + x^3) \cdot \mathbf{g}(x)$
(1 0 1 1)	1 1 1 1 1 1 1	$1 + x + x^2 + x^3 + x^4 + x^5 + x^6 = (1 + x^2 + x^3) \cdot \mathbf{g}(x)$
(1 1 1 1)	1 0 0 1 0 1 1	$1 + x^3 + x^5 + x^6 = (1 + x + x^2 + x^3) \cdot \mathbf{g}(x)$

- Consider the polynomial $x\mathbf{g}(x), x^2\mathbf{g}(x), \dots, x^{n-r-1}\mathbf{g}(x)$. They are cyclic shifts of $\mathbf{g}(x)$ and hence code polynomials in $\mathbf{C}$.

Since $\mathbf{C}$ is linear, a linear combination of

$$\mathbf{g}(x), x\mathbf{g}(x), \dots, x^{n-r-1}\mathbf{g}(x),$$

$$\begin{aligned} \mathbf{v}(x) &= u_0\mathbf{g}(x) + u_1x\mathbf{g}(x) + \dots + u_{n-r-1}x^{n-r-1}\mathbf{g}(x) \\ &= (u_0 + u_1x + \dots + u_{n-r-1}x^{n-r-1})\mathbf{g}(x), \end{aligned}$$

is also a code polynomial where $u_i \in \{0, 1\}$.

- Let $\mathbf{g}(x) = 1 + g_1x + \dots + g_{r-1}x^{r-1} + x^r$ be the nonzero code polynomial of minimum degree in an (n, k) cyclic code $\mathbf{C}$. A binary polynomial of degree $n - 1$ or less is a code polynomial *if and only if* it is a multiple of $\mathbf{g}(x)$.

Proof: Let $\mathbf{v}(x)$ be a binary polynomial of degree $n - 1$ or less. Suppose that $\mathbf{v}(x)$ is a multiple of $\mathbf{g}(x)$. Then

$$\mathbf{v}(x) = (a_0 + a_1x + \dots + a_{n-r-1}x^{n-r-1})\mathbf{g}(x)$$

$$= a_0 \mathbf{g}(x) + a_1 x \mathbf{g}(x) + \cdots + a_{n-r-1} x^{n-r-1} \mathbf{g}(x).$$

Since $\mathbf{v}(x)$ is a linear combination of the code polynomials, $\mathbf{g}(x), x\mathbf{g}(x), \dots, x^{n-r-1}\mathbf{g}(x)$, it is a code polynomial in $\mathbf{C}$.

Let $\mathbf{v}(x)$ be a code polynomial in $\mathbf{C}$. Dividing $\mathbf{v}(x)$ by $\mathbf{g}(x)$, we obtain

$$\mathbf{v}(x) = \mathbf{a}(x)\mathbf{g}(x) + \mathbf{b}(x),$$

where the degree of $\mathbf{b}(x)$ is less than the degree of $\mathbf{g}(x)$. Since $\mathbf{v}(x)$ and $\mathbf{a}(x)\mathbf{g}(x)$ are code polynomials, $\mathbf{b}(x)$ is also a code polynomial. Suppose $\mathbf{b}(x) \neq 0$. Then $\mathbf{b}(x)$ is a code polynomial with less degree than that of $\mathbf{g}(x)$. Contradiction.

- The number of binary polynomials of degree $n - 1$ or less that are multiples of $\mathbf{g}(x)$ is 2^{n-r} .
- There are total of 2^k code polynomials in $\mathbf{C}$, $2^{n-r} = 2^k$, i.e., $r = n - k$.

- The polynomial $\mathbf{g}(x)$ is called the *generator polynomial* of the code.
- The degree of $\mathbf{g}(x)$ is equal to the number of parity-check digits of the code.
- The generator polynomial $\mathbf{g}(x)$ of an (n, k) cyclic code is a factor of $x^n + 1$.

Proof: We have

$$x^k \mathbf{g}(x) = (x^n + 1) + \mathbf{g}^{(k)}(x).$$

Since $\mathbf{g}^{(k)}(x)$ is the code polynomial obtained by shifting $\mathbf{g}(x)$ to the right cyclically k times, $\mathbf{g}^{(k)}(x)$ is a multiple of $\mathbf{g}(x)$.

Hence,

$$x^n + 1 = \{x^k + \mathbf{a}(x)\} \mathbf{g}(x).$$

- If $\mathbf{g}(x)$ is a polynomial of degree $n - k$ and is a factor of $x^n + 1$, then $\mathbf{g}(x)$ generates an (n, k) cyclic code.

Proof: A linear combination of $\mathbf{g}(x), x\mathbf{g}(x), \dots, x^{k-1}\mathbf{g}(x)$,

$$\begin{aligned}\mathbf{v}(x) &= a_0\mathbf{g}(x) + a_1x\mathbf{g}(x) + \cdots + a_{k-1}x^{k-1}\mathbf{g}(x) \\ &= (a_0 + a_1x + \cdots + a_{k-1}x^{k-1})\mathbf{g}(x),\end{aligned}$$

is a polynomial of degree $n - 1$ or less and is a multiple of $\mathbf{g}(x)$. There are a total of 2^k such polynomials, and they form an (n, k) linear code.

Let $\mathbf{v}(x) = v_0 + v_1x + \cdots + v_{n-1}x^{n-1}$ be a code polynomial in this code. We have

$$\begin{aligned}x\mathbf{v}(x) &= v_0x + v_1x^2 + \cdots + v_{n-1}x^n \\ &= v_{n-1}(x^n + 1) + (v_{n-1} + v_0x + \cdots + v_{n-2}x^{n-1}) \\ &= v_{n-1}(x^n + 1) + \mathbf{v}^{(1)}(x).\end{aligned}$$

Since both $x\mathbf{v}(x)$ and $x^n + 1$ are divisible by $\mathbf{g}(x)$, $\mathbf{v}^{(1)}$ must be divisible by $\mathbf{g}(x)$. Hence, $\mathbf{v}^{(1)}(x)$ is a code polynomial and the code generated by $\mathbf{g}(x)$ is a cyclic code.

- Suppose that the message to be encoded is

$\mathbf{u} = (u_0, u_1, \dots, u_{k-1})$. Then

$$x^{n-k}\mathbf{u}(x) = u_0x^{n-k} + u_1x^{n-k+1} + \dots + u_{k-1}x^{n-1}.$$

Dividing $x^{n-k}\mathbf{u}(x)$ by $\mathbf{g}(x)$, we have

$$x^{n-k}\mathbf{u}(x) = \mathbf{a}(x)\mathbf{g}(x) + \mathbf{b}(x).$$

Since the degree of $\mathbf{g}(x)$ is $n - k$, the degree of $\mathbf{b}(x)$ must be $n - k - 1$ or less. Then

$$\mathbf{b}(x) + x^{n-k}\mathbf{u}(x) = \mathbf{a}(x)\mathbf{g}(x)$$

is a multiple of $\mathbf{g}(x)$ and therefore it is a code polynomial.

$$\begin{aligned} \mathbf{b}(x) + x^{n-k}\mathbf{u}(x) &= b_0 + b_1x + \dots + b_{n-k-1}x^{n-k-1} \\ &\quad + u_0x^{n-k} + u_1x^{n-k+1} + \dots + u_{k-1}x^{n-1} \end{aligned}$$

then corresponds to the codeword

$$(b_0, b_1, \dots, b_{n-k-1}, u_0, u_1, \dots, u_{k-1}).$$

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(1 1 0 0)	1 0 1 1 1 0 0	$1 + x^2 + x^3 + x^4 = (1 + x) \cdot \mathbf{g}(x)$
(0 0 1 0)	0 0 1 1 0 1 0	$x^2 + x^3 + x^5 = x^2 \cdot \mathbf{g}(x)$
(1 0 1 0)	1 1 1 0 0 1 0	$1 + x + x^2 + x^5 = (1 + x^2) \cdot \mathbf{g}(x)$
(0 1 1 0)	0 1 0 1 1 1 0	$x + x^3 + x^4 + x^5 = (x + x^2) \cdot \mathbf{g}(x)$
(1 1 1 0)	1 0 0 0 1 1 0	$1 + x^4 + x^5 = (1 + x + x^2) \cdot \mathbf{g}(x)$
(0 0 0 1)	0 0 0 1 1 0 1	$x^3 + x^4 + x^6 = x^3 \cdot \mathbf{g}(x)$
(1 0 0 1)	1 1 0 0 1 0 1	$1 + x + x^4 + x^6 = (1 + x^3) \cdot \mathbf{g}(x)$
(0 1 0 1)	0 1 1 1 0 0 1	$x + x^2 + x^3 + x^6 = (x + x^3) \cdot \mathbf{g}(x)$
(1 1 0 1)	1 0 1 0 0 0 1	$1 + x^2 + x^6 = (1 + x + x^3) \cdot \mathbf{g}(x)$
(0 0 1 1)	0 0 1 0 1 1 1	$x^2 + x^4 + x^5 + x^6 = (x^2 + x^3) \cdot \mathbf{g}(x)$
(1 0 1 1)	1 1 1 1 1 1 1	$1 + x + x^2 + x^3 + x^4 + x^5 + x^6 = (1 + x^2 + x^3) \cdot \mathbf{g}(x)$
(1 1 1 1)	1 0 0 1 0 1 1	$1 + x^3 + x^5 + x^6 = (1 + x + x^2 + x^3) \cdot \mathbf{g}(x)$

Generator and Parity-Check Matrices

- The generator matrix of an (n, k) code $\mathbf{C}$ is as follows:

$$\mathbf{G} = \begin{bmatrix} g_0 & g_1 & g_2 & \cdot & \cdot & \cdot & \cdot & \cdot & g_{n-k} & 0 & 0 & 0 & \cdot & \cdot & 0 \\ 0 & g_0 & g_1 & g_2 & \cdot & \cdot & \cdot & \cdot & \cdot & g_{n-k} & 0 & 0 & \cdot & \cdot & 0 \\ 0 & 0 & g_0 & g_1 & g_2 & \cdot & \cdot & \cdot & \cdot & \cdot & g_{n-k} & 0 & \cdot & \cdot & 0 \\ \cdot & & & & & & & & & & & & & & \cdot \\ \cdot & & & & & & & & & & & & & & \cdot \\ \cdot & & & & & & & & & & & & & & \cdot \\ 0 & 0 & \cdot & \cdot & \cdot & 0 & g_0 & g_1 & g_2 & \cdot & \cdot & \cdot & \cdot & \cdot & g_{n-k} \end{bmatrix}$$

- In general, $\mathbf{G}$ is not in systematic form. However, it can be put into systematic form with row operation.
- Let

$$x^n + 1 = \mathbf{g}(x)\mathbf{h}(x),$$

where the polynomial $\mathbf{h}(x)$ has the degree k and is of the following form:

$$\mathbf{h}(x) = h_0 + h_1x + \cdots + h_kx^k$$

with $h_0 = h_k = 1$.

- A parity-check matrix of $\mathbf{C}$ may be obtained from $\mathbf{h}(x)$.
- Let $\mathbf{v}$ be a codeword in $\mathbf{C}$ and $\mathbf{v}(x) = \mathbf{a}(x)\mathbf{g}(x)$. Then

$$\begin{aligned}\mathbf{v}(x)\mathbf{h}(x) &= \mathbf{a}(x)\mathbf{g}(x)\mathbf{h}(x) \\ &= \mathbf{a}(x)(x^n + 1) \\ &= \mathbf{a}(x) + x^n\mathbf{a}(x).\end{aligned}$$

Since the degree of $\mathbf{a}(x)$ is $k - 1$ or less, the powers $x^k, x^{k+1}, \dots, x^{n-1}$ do not appear in $\mathbf{a}(x) + x^n\mathbf{a}(x)$. Therefore,

$$\sum_{i=0}^k h_i v_{n-i-j} = 0 \text{ for } 1 \leq j \leq n - k.$$

We take the *reciprocal* of $\mathbf{h}(x)$,

$$x^k \mathbf{h}(x^{-1}) = h_k + h_{k-1}x + h_{k-2}x^2 + \cdots + h_0x^k,$$

and can see that $x^k \mathbf{h}(x^{-1})$ is also a factor of $x^n + 1$. $x^k \mathbf{h}(x^{-1})$ then generates an $(n, n - k)$ cyclic code with the following $(n - k) \times n$ matrix as a generator matrix:

$$\mathbf{H} = \begin{bmatrix} h_k & h_{k-1} & h_{k-2} & \cdot & \cdot & \cdot & \cdot & h_0 & 0 & 0 & \cdot & \cdot & 0 \\ 0 & h_k & h_{k-1} & h_{k-2} & \cdot & \cdot & \cdot & \cdot & h_0 & 0 & \cdot & \cdot & 0 \\ 0 & 0 & h_k & h_{k-1} & \cdot & \cdot & \cdot & \cdot & \cdot & h_0 & \cdot & \cdot & 0 \\ \cdot & & & & & & & & & & & & \cdot \\ \cdot & & & & & & & & & & & & \cdot \\ \cdot & & & & & & & & & & & & \cdot \\ 0 & 0 & \cdot & \cdot & 0 & h_k & h_{k-1} & h_{k-2} & \cdot & \cdot & \cdot & \cdot & h_0 \end{bmatrix}$$

Then $\mathbf{H}$ is a parity-check matrix of the cyclic code $\mathbf{C}$. We call $\mathbf{h}(x)$ the *parity polynomial* of $\mathbf{C}$.

- Let $\mathbf{C}$ be an (n, k) cyclic code with generator polynomial $\mathbf{g}(x)$. The dual code of $\mathbf{C}$ is also cyclic and is generated by the polynomial $x^k \mathbf{h}(x^{-1})$, where $\mathbf{h}(x) = (x^n + 1)/\mathbf{g}(x)$.
- Let

$$x^{n-k+i} = \mathbf{a}_i(x)\mathbf{g}(x) + \mathbf{b}_i(x) \text{ for } 0 \leq i \leq k - 1,$$

where $\mathbf{b}_i(x) = b_{i0} + b_{i1}x + \cdots + b_{i(n-k-1)}x^{n-k-1}$. Since $\mathbf{b}_i(x) + x^{n-k+i}$ are multiples of $\mathbf{g}(x)$, they are code polynomials. Then

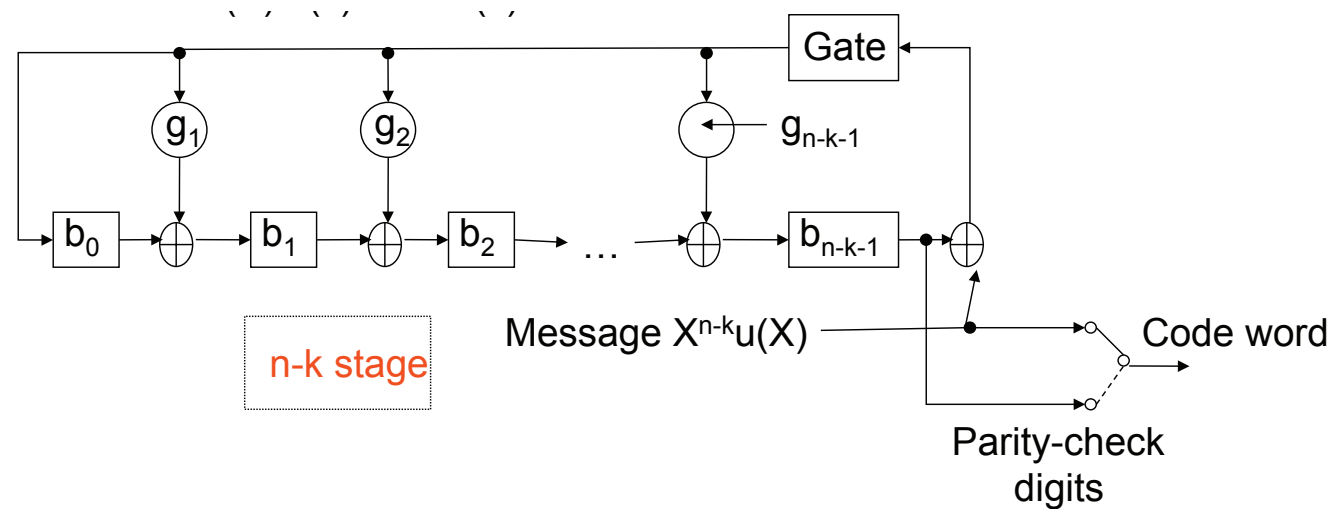
$$\mathbf{G} = \begin{bmatrix} b_{00} & b_{01} & b_{02} & \cdots & b_{0(n-k-1)} & 1 & 0 & 0 & \cdots & 0 \\ b_{10} & b_{11} & b_{12} & \cdots & b_{1(n-k-1)} & 0 & 1 & 0 & \cdots & 0 \\ b_{20} & b_{21} & b_{22} & \cdots & b_{2(n-k-1)} & 0 & & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ b_{(k-1)0} & b_{(k-1)1} & b_{(k-1)2} & \cdots & b_{(k-1)(n-k-1)} & 0 & 0 & 0 & \cdots & 1 \end{bmatrix}.$$

- The corresponding parity-check matrix for $\mathbf{C}$ is

$$\mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & b_{00} & b_{10} & b_{20} & \cdots & b_{(k-1)0} \\ 0 & 1 & 0 & \cdots & 0 & b_{01} & b_{11} & b_{21} & \cdots & b_{(k-1)1} \\ 0 & 0 & 1 & \cdots & 0 & b_{02} & b_{12} & b_{22} & \cdots & b_{(k-1)2} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & b_{0(n-k-1)} & b_{1(n-k-1)} & b_{2(n-k-1)} & \cdots & b_{(k-1)(n-k-1)} \end{bmatrix}.$$

Encoding of Cyclic Codes

- Encoding process: (1) Multiply $\mathbf{u}(x)$ by x^{n-k} ; (2) divide $x^{n-k}\mathbf{u}(x)$ by $\mathbf{g}(x)$; (3) form the codeword $\mathbf{b}(x) + x^{n-k}\mathbf{u}(x)$.

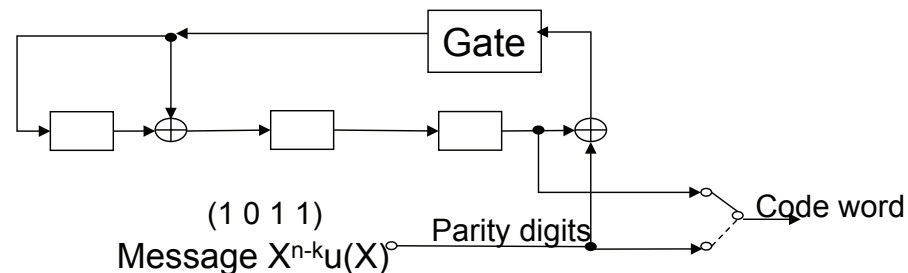


Example

- Consider the $(7, 4)$ cyclic code generated by $g(x) = 1 + x + x^3$. Suppose the message $u = (1\ 0\ 1\ 1)$ is to be encoded. The contents in the register are as follows:

Input	Register contents
	0 0 0 (initial state)
1	1 1 0 (first shift)
1	1 0 1 (second shift)
0	1 0 0 (third shift)
1	<u>1 0 0</u> (fourth shift)

After four shifts, the contents of the register are $(1\ 0\ 0)$. Thus the complete codeword is $(1\ 0\ 0\ 1\ 0\ 1\ 1)$.



Encoding by Parity Polynomial

- Since $h_k = 1$, we have

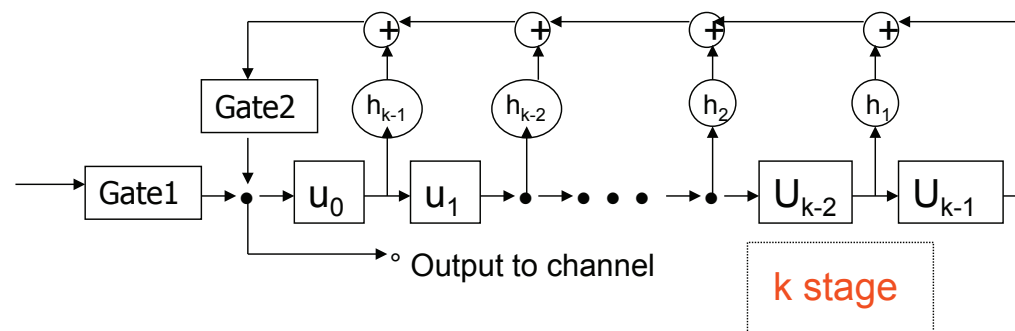
$$v_{n-k-j} = \sum_{i=0}^{k-1} h_i v_{n-i-j} \text{ for } 1 \leq j \leq n-k,$$

which is known as a *difference equation*.

$$v_{n-k-1} = h_0 v_{n-1} + h_1 v_{n-2} + \cdots + h_{k-1} v_{n-k} = u_{k-1} + h_1 u_{k-2} + \cdots + h_{k-1} u_0$$

$$v_{n-k-2} = u_{k-2} + h_1 u_{k-3} + \cdots + h_{k-1} v_{n-k-1}$$

- Encoding circuit:



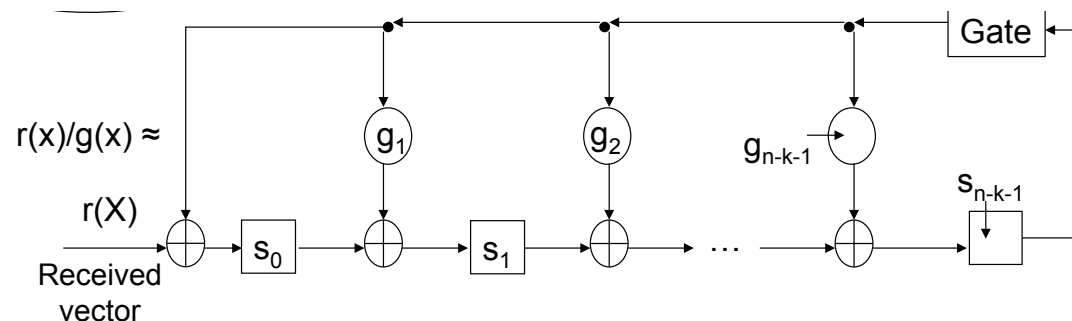
The codeword that corresponds to the message $(1\ 0\ 1\ 1)$ is $(1\ 0\ 0\ 1\ 0\ 1\ 1)$.

Syndrome Computation

- Let $\mathbf{r} = (r_0, r_1, \dots, r_{n-1})$ be the received vector. The *syndrome* is calculated as $\mathbf{s} = \mathbf{r} \cdot \mathbf{H}^T$, where $\mathbf{H}$ is the parity-check matrix.
- If the syndrome is not identical to zero, $\mathbf{r}$ is not a codeword, and the presence of errors has been detected.
- Dividing $\mathbf{r}(x)$ by the generator polynomial $\mathbf{g}(x)$, we obtain

$$\mathbf{r}(x) = \mathbf{a}(x)\mathbf{g}(x) + \mathbf{s}(x).$$

- The $n - k$ coefficients of $\mathbf{s}(x)$ form the syndrome $\mathbf{s}$. We call $\mathbf{s}(x)$ the *syndrome*.



- If $\mathbf{C}$ is a systematic code, then the syndrome is simply the vector sum of the received parity digits, and the parity-check digits recomputed from the received information digits.
- Let $\mathbf{s}(x)$ be the syndrome of a received polynomial $\mathbf{r}(x)$. Then the remainder $\mathbf{s}^{(1)}(x)$ resulting from dividing $x\mathbf{s}(x)$ by the generator polynomial $\mathbf{g}(x)$ is the syndrome of $\mathbf{r}^{(1)}(x)$, which is a cyclic shift of $\mathbf{r}(x)$.

Proof: We have

$$x\mathbf{r}(x) = r_{n-1}(x^n + 1) + \mathbf{r}^{(1)}(x).$$

Then

$$\mathbf{c}(x)\mathbf{g}(x) + \boldsymbol{\rho}(x) = r_{n-1}\mathbf{g}(x)\mathbf{h}(x) + x[\mathbf{a}(x)\mathbf{g}(x) + \mathbf{s}(x)],$$

where $\boldsymbol{\rho}(x)$ is the remainder resulting from dividing $\mathbf{r}^{(1)}(x)$ by $\mathbf{g}(x)$. Then $\boldsymbol{\rho}(x)$ is the syndrome of $\mathbf{r}^{(1)}(x)$. Rearranging the

above equation, we have

$$x\mathbf{s}(x) = [\mathbf{c}(x) + r_{n-1}\mathbf{h}(x) + x\mathbf{a}(x)]\mathbf{g}(x) + \boldsymbol{\rho}(x).$$

It is clearly that $\boldsymbol{\rho}(x)$ is also the remainder resulting from dividing $x\mathbf{s}(x)$ by $\mathbf{g}(x)$. Therefore, $\boldsymbol{\rho}(x) = \mathbf{s}^{(1)}(x)$.

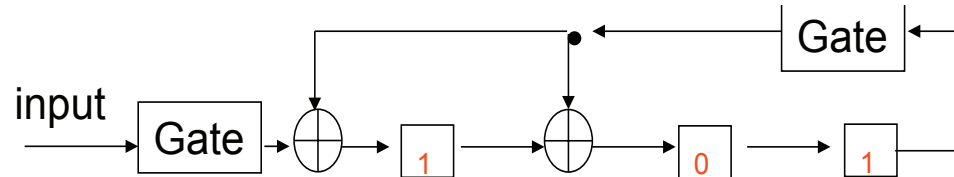
- The remainder $\mathbf{s}^{(i)}(x)$ resulting from dividing $x^i\mathbf{s}(x)$ by the generator polynomial $\mathbf{g}(x)$ is the syndrome of $\mathbf{r}^{(i)}(x)$, which is the i th cyclic shift of $\mathbf{r}(x)$.

Example

Consider the $(7, 4)$ cyclic code generated by $g(x) = 1 + x + x^3$.

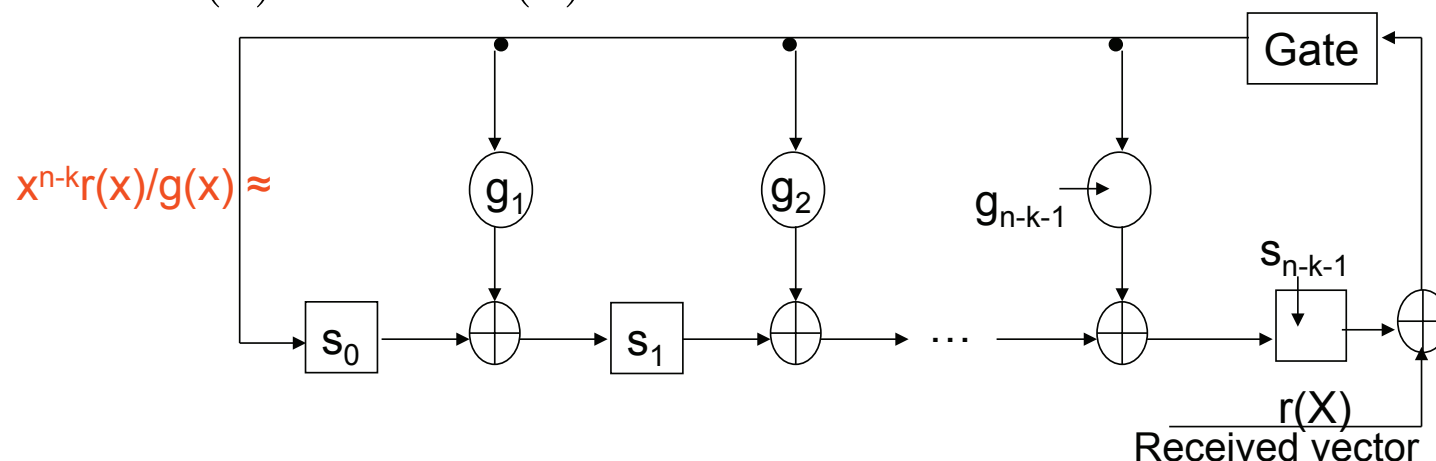
Suppose that the received vector is $\mathbf{r} = (0 \ 0 \ 1 \ 0 \ 1 \ 1 \ 0)$. The syndrome of $\mathbf{r}$ is $\mathbf{s} = (1 \ 0 \ 1)$. As the received vector is shifted into the circuit, the contents in the register are as follows:

Shift	Input	Register contents
		0 0 0 (initial state)
1	0	0 0 0
2	1	1 0 0
3	1	1 1 0
4	0	0 1 1
5	1	0 1 1
6	0	1 1 1
7	0	1 0 1 (syndrome $\mathbf{s}$)
8	-	1 0 0 (syndrome $\mathbf{s}^{(1)}$)
9	-	0 1 0 (syndrome $\mathbf{s}^{(2)}$)



If the register is shifted once more with the input gate disabled, the new contents will be $\mathbf{s}^{(1)} = (1 \ 0 \ 0)$, which is the syndrome of $\mathbf{r}^{(1)} = (0 \ 0 \ 0 \ 1 \ 0 \ 1 \ 1)$.

- We may shift the received vector $\mathbf{r}(x)$ into the syndrome register from the right end. However, after the entire $\mathbf{r}(x)$ has been shifted into the register, the contents in the register do not form the syndrome of $\mathbf{r}(x)$; rather, they form the syndrome $\mathbf{s}^{(n-k)}(x)$ of $\mathbf{r}^{(n-k)}(x)$.



Proof: We have

$$x^{n-k} \mathbf{r}(x) = \mathbf{a}(x) \mathbf{g}(x) + \boldsymbol{\rho}(x).$$

It is known that

$$x^{n-k} \mathbf{r}(x) = \mathbf{b}(x)(x^n + 1) + \mathbf{r}^{(n-k)}(x).$$

Hence,

$$\mathbf{r}^{(n-k)}(x) = [\mathbf{b}(x)\mathbf{h}(x) + \mathbf{a}(x)]\mathbf{g}(x) + \boldsymbol{\rho}(x).$$

When $\mathbf{r}^{(n-k)}(x)$ is divided by $\mathbf{g}(x)$, $\boldsymbol{\rho}(x)$ is also the remainder.
Therefore, $\boldsymbol{\rho}(x)$ is indeed the syndrome of $\mathbf{r}^{(n-k)}(x)$.

Error Detection

- Let $\mathbf{v}(x)$ be the transmitted codeword and $\mathbf{e}(x) = e_0 + e_1x + \cdots + e_{n-1}x^{n-1}$ be the error pattern. Then

$$\mathbf{r}(x) = \mathbf{v}(x) + \mathbf{e}(x) = \mathbf{b}(x)\mathbf{g}(x) + \mathbf{e}(x).$$

- Following the definition of the syndrome, we have

$$\mathbf{e}(x) = [\mathbf{a}(x) + \mathbf{b}(x)]\mathbf{g}(x) + \mathbf{s}(x).$$

This shows that the syndrome is equal to the remainder resulting from dividing the error pattern by the generator polynomial.

- The decoder has to estimate $\mathbf{e}(x)$ based on the syndrome $\mathbf{s}(x)$.
- If $\mathbf{e}(x)$ is identical to a codeword, $\mathbf{e}(x)$ is an undetectable error pattern.
- The error-detection circuit is simply a syndrome circuit with an

OR gate with the syndrome digits as inputs.

- For a cyclic code, an error pattern with errors confined to i high-order positions and $\ell - i$ low-order positions is also regarded as a burst of length ℓ or less. Such a burst is called *end-around* burst.
- An (n, k) cyclic code can detect any error burst of length $n - k$ or less, including the end-around bursts.

Proof: Suppose that the error pattern is a burst of the length of $n - k$ or less. Then

$$\mathbf{e}(x) = x^j \mathbf{B}(x),$$

where $0 \leq j \leq n - 1$ and $\mathbf{B}(x)$ is a polynomial of degree $n - k - 1$ or less. Since the degree of $\mathbf{B}(x)$ is less than that of $\mathbf{g}(x)$, $\mathbf{B}(x)$ is not divisible by $\mathbf{g}(x)$. Since $\mathbf{g}(x)$ is a factor of $x^n + 1$ and x is not a factor of $\mathbf{g}(x)$, $\mathbf{g}(x)$ and x^j must be relatively prime. Therefore, $\mathbf{e}(x)$ is not divisible by $\mathbf{g}(x)$. The

last part of the above statement is left as an exercise.

- The fraction of undetectable bursts of length $n - k + 1$ is $2^{-(n-k-1)}$.

Proof: Consider the bursts of length $n - k + 1$ starting from the i th digit position and ending at the $(i + n - k)$ th digit position. There are 2^{n-k-1} such bursts. Among these bursts, the only one that cannot be detected is

$$e(x) = x^i g(x).$$

Therefore, the fraction of undetectable bursts of length $n - k + 1$ starting from the i th digit position is $2^{-(n-k-1)}$.

- For $\ell > n - k + 1$, the fraction of undetectable error bursts of length ℓ is $2^{-(n-k)}$.